ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | 979-8-3503-6874-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICASSP49660.2025.10888452

Audio Features Investigation for Singing Voice Deepfake Detection

Mahyar Gohari
University of Brescia
Brescia, Italy
m.goharimoghaddam@unibs.it

Davide Salvi
Politecnico di Milano
Milano, Italy
davide.salvi@polimi.it

Paolo Bestagini
Politecnico di Milano
Milano, Italy
paolo.bestagini@polimi.it

Nicola Adami
University of Brescia
Brescia, Italy
nicola.adami@unibs.it

***Abstract*—The audio forensics field has recently faced a new challenge: singing voice deepfake detection. Current approaches to tackle this problem have borrowed methods initially developed for the more established task of speech deepfake detection, often simply retraining these systems on singing voice data. However, effective speech detection techniques may not necessarily perform well on singing voice, and there has been limited research on identifying the factors that can improve detection specifically in the singing domain. This paper investigates the effectiveness of various audio representations and features for discriminating real and synthetically generated singing voice signals. We evaluate two Convolutional Neural Network (CNN)-based detection systems using a wide range of audio representations, including handcrafted, learning-based, and pre-trained features. Through a systematic analysis, we aim to understand the key factors that can improve the performance of deepfake detection methods for singing voices. Additionally, we investigate the differences between singing voice and speech detection, highlighting the implications of the feature sets considered. Our results offer valuable insights and guidance for developing more advanced and effective singing voice deepfake detection systems in the future.**

***Index Terms*—Singing voice, deepfake, audio forensics**

I. Introduction

Recent advancements in speech generative models have introduced new challenges in audio forensics and content authenticity verification, creating unexplored scenarios and problems that need to be addressed. One of these is Singing Voice Deepfake Detection (SVDD). Singing voice deepfakes enable the impersonation or cloning of any individual's voice, making them sing any song or melody. Although this might initially appear to be an entertaining application, it can have severe implications for both personal and commercial security. There have already been cases where songs featuring the cloned voices of famous singers have gone viral without the artists' consent, resulting in image damage and copyright violations [1].This has prompted the scientific community to develop new methods for preventing malicious consequences and detecting singing voice deepfakes [2], [3].

Most of the methods considered to tackle this task have been heavily inspired by systems used for speech deepfake detection, given the clear closeness of the two problems. However, while synthetic speech detection has seen considerable progress and several methods have been developed specifically to exploit some speech nuances [4]–[8], limited investigation has been done to design systems tailored to the singing voice, considering, for instance, the unique characteristics of singing, such as pitch variation, timbre, and expressiveness. One of the few examples in this regard is SingGraph [9], which combines an acoustic understanding model with a speech SSL model to simultaneously analyze the music and voice for SVDD. On the other hand, the authors of [10] propose to adapt techniques originally developed for synthetic speech detection (i.e., AASIST [11], ResNet [12] and Wav2Vec 2.0 [13]) to the problem of synthetic singing voice detection. In doing so, they highlight the differences obtained when training on plain speech and testing on singing voices or vice versa. However, despite this effort, the specific task of detecting synthetic singing voices remains relatively unexplored.

A significant gap in the literature lies in understanding which audio features or representations are most effective at distinguishing between real and synthetic singing voices. While a similar study has been conducted for speech [14], i.e., identifying the most valuable feature set to perform the detection task, such an analysis has not yet been performed for singing voices. This paper aims to fill this gap by systematically evaluating a range of audio representations within two different Convolutional Neural Network (CNN)-based pipelines for SVDD. Rather than proposing a new detection model, our primary goal is to gain insights into the efficacy of different audio features in capturing the traces left by singing voice synthesizers. We analyze the performance of two distinct CNN-based classifiers across various audio features as input, aiming to determine which representations are most informative for the detection task. We consider 10 different audio features, including hand-crafted, learned, and pre-trained representations, aiming to make our analysis as comprehensive as possible. We repeat the same study considering both plain speech and singing voices to explore the differences between these two domains. Our goal is to determine whether the same feature sets can effectively address both tasks or if different representations are needed for each. A feature set that performs well for detecting synthetic speech may not be as effective for identifying synthetic singing voices, or vice versa. We

This work is supported by the European Union under the Italian National Recovery and Resilience Plan (NRRP) of NextGenerationEU (PE00000001 - program "RESTART", PE00000014 - program "SERICS"). This work is supported by the FOSTERER project, funded by the Italian Ministry of University, and Research within the PRIN 2022 program.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on February 28,2026 at 09:53:11 UTC from IEEE Xplore. Restrictions apply.

also conducted the same tests with two different detectors to determine whether high detection performance is mainly due to the choice of specific feature sets or if the results are affected by the specific architecture used. This research not only deepens our understanding of the problem but also provides valuable guidance for developing more effective detection systems for SVDD in future works.

II. Proposed Method

In this paper, we consider the problem of Singing Voice Deepfake Detection (SVDD) and perform a systematic analysis to understand which is the most effective audio feature set to address this task. During our analysis, we consider both singing voice and plain speech signals, aiming to identify the most relevant features for SVDD while also exploring the differences between these two domains. The insights gained from this investigation are intended to inform the development of more effective singing voice detectors in the future.

The motivation for this study stems from the fact that, for plain speech, extensive research has already identified the most suitable feature sets for deepfake detection [15], [16]. However, given the novelty of the SVDD task, such an analysis has not yet been conducted for singing voices. Consequently, current SVDD methods often rely on approaches introduced for synthetic speech detection, which may not be optimal. These methods do not leverage specific characteristics unique to singing voices, which could enhance detection performance, as these vocal aspects are also the most challenging to replicate when generating synthetic singing voice signals [17], [18].

We tackle this problem by considering a CNN-based model trained to distinguish whether an input audio signal containing a singing voice is real (i.e., performed by a human) or synthetic (i.e., generated by an algorithm). As the model can accept different audio representations as input, we examine how its performance is influenced by using different feature sets.

The problem we address can be formally defined as follows. Let us consider a discrete-time input signal $\mathbf{x}$, sampled with a sampling frequency f_s and containing a singing voice. We consider $\mathbf{x}$ as associated with a class $y \quad [0\ 1]$, which indicates whether the signal is real ($y = 0$) or has been synthetically generated ($y = 1$). The goal of the singing voice deepfake detection task is to develop a detection model $\mathcal{M}$ that takes $\mathbf{x}$ as input and outputs an estimate of its class as

$$y = \mathcal{M}(\mathbf{x}) \tag{1}$$

where y is a continuous value in the range $[0\ 1]$ indicating the likelihood that the signal $\mathbf{x}$ is fake.

Inspired by [14], we consider the detection pipeline of $\mathcal{M}$ as divided into two steps, as shown in Figure 1. The first step is the *feature extraction*, followed by the *classification*. During *feature extraction*, the input audio signal is transformed into a feature vector that captures its essential characteristics. Formally, we define the extractor $\mathcal{E}$ such that $\mathbf{f} = \mathcal{E}(\mathbf{x})$, where $\mathbf{f}$ represents the extracted feature vector. In the *classification* step, the feature vector is fed to a CNN-based model, which processes it to perform binary classification, outputting the likelihood that the input signal is synthetic. Formally, we define the classifier $\mathcal{C}$ such that $y = \mathcal{C}(\mathbf{f})$ represents the estimated likelihood that the singing voice signal $\mathbf{x}$ is fake.

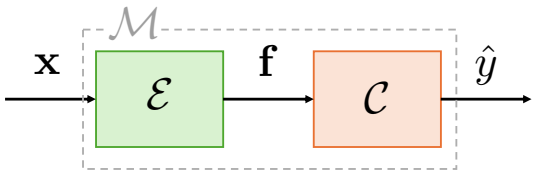

Fig. 1. Considered pipeline for the singing voice deepfake detection model $\mathcal{M}$. The extractor $\mathcal{E}$ computes features $\mathbf{f}$ from the input audio $\mathbf{x}$. Features are fed to the classifier $\mathcal{C}$ that estimates the fake singing likelihood $\hat{y}$.

The overall model $\mathcal{M}$ is obtained by the combination of the feature extractor and the classifier as

$$y = \mathcal{M}(\mathbf{x}) = \mathcal{C}(\mathcal{E}(\mathbf{x})) \tag{2}$$

Our goal is to investigate how the performance of $\mathcal{M}$ varies when different feature extractors $\mathcal{E}$ are used.

To ensure our findings are generalizable and not limited to a single model, we conduct this analysis using two different CNN-based detectors. Additionally, we conduct this analysis for both singing voices and plain speech, emphasizing the differences between the best-performing feature sets for each domain. In the following section, we provide all the details related to the used features and classification models.

III. Experimental Setup

In this section, we present all the details of our experimental campaign. We start by introducing the audio features we analyzed, and then we detail the used datasets. Finally, we describe the considered models and the setup we used to train and test them.

A. Audio Features

As discussed in Section II, our analysis includes various feature sets, ranging from traditional hand-crafted spectral features to more complex representations, such as learned filterbanks and pre-trained models.

Hand-crafted features. These audio representations are highly recognized in signal processing, and their use in the speech deepfake detection field has been widely investigated. These features are valued for their capability to offer compact yet insightful representations of an audio signal's spectrum. In our analysis, we considered five different feature sets, each computed using a window length of $25\,\mathrm{ms}$ and a hop size of $10\,\mathrm{ms}$. The feature sets include: *log-spectrograms*, which represent the logarithm of the audio signal's power spectrum; *mel-spectrograms*, which use an 80-band mel-filterbank to capture frequency components; *Linear Frequency Cepstral Coefficient (LFCC)*, extracted with 40 coefficients along with their first and second-order derivatives; *Mel Frequency Cepstral Coefficient (MFCC)*, computed using a mel-filterbank and the same configuration as LFCCs; and the *Constant-Q Transform (CQT)*, which is calculated with 84 bins and 12 bins per octave. These diverse feature sets are designed to capture

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on February 28,2026 at 09:53:11 UTC from IEEE Xplore. Restrictions apply.

different aspects of the audio signal's characteristics, and each may lead to varying results in audio deepfake detection.

Learning-based features. We utilize two different learnable acoustic front-ends that update the weights of their filter banks during the detectors' training process to better address the specific task. These types of features have already demonstrated strong performance in speech deepfake detection [11], [19]. For our study, we employ *LEAF* [20] and *SincConv* [21], computed with 40 and 64 filters respectively.

Pretrained features. These feature sets are generated from models pre-trained on large datasets across multiple speech tasks, providing high-level audio representations that proved to be effective in speech emotion recognition [22], voice disorder detection [23] and speech deepfake detection [24]. We include features computed with *Whisper* tiny [25], *WavLM* base [26], with the base model, and *Wav2Vec 2.0* XLS-R 300M [27].

B. Classifiers

We employ two CNN-based classifiers to evaluate the contributions of various audio features for singing voice deepfake detection. We do so to provide results that can be generalized across multiple detectors, thereby making our findings more relevant and beneficial for the development of new singing voice deepfake detectors in the future.

ResNet-18 [28]. This CNN model encompasses 18 layers that exploit residual connections to effectively address the vanishing gradient problem. We considered the version of this model pre-trained on ImageNet [29], as it is known for its efficiency in image classification and has often been adapted to audio classification tasks [30]. This model is augmented with a dropout layer [31] preceding two fully connected layers to enhance its generalization capabilities. To accommodate audio features, which are typically 2D, as opposed to the 3D RGB images, we modified the initial layer of ResNet-18 by applying channel-wise averaging.

LCNN [32]. This is a Light CNN model adapted to the speech deepfake detection task. It operates on 2D features and integrates several convolutional layers with varying kernel sizes and strides to capture different levels of abstraction, together with multi-level feature mapping techniques to enhance feature extraction. It has been considered as baseline in the ASVspoof 2021 challenge, for the speech deepfake detection task [33].

C. Datasets

We considered two different datasets to compare the results we obtained on singing voice deepfake detection with those obtained with speech deepfake detection.

CtrSVDD [34]. This dataset has been introduced in the SVDD 2024 Challenge and provides a comprehensive framework for evaluating singing voice deepfake detection models. The dataset contains a diverse range of both real and deepfake vocals, generated using diverse synthesis methods. The bonafide vocals in the dataset come from several open-access singing voice collections, such as Opencpop [35], M4Singer [36], Kising [37], and Japanese datasets such as Kiritan [38] and JVS-MuSiC [39]. The deepfake vocals are created using 14 distinct synthesis methods, which include both Singing Voice Synthesis (SVS) and Singing Voice Conversion (SVC) models. The dataset is divided into training, development, and evaluation subsets to ensure a diverse representation of singers and synthesis methods across partitions, minimizing potential biases in model evaluation.

ASVspoof 2019 [40]. This speech deepfake detection dataset has been released for the homonymous challenge, and was designed to develop effective ASV models. It includes real speech from the VCTK corpus [41] and synthetic speech generated by 19 different algorithms. As in the case of CtrSVDD, this corpus is divided into three subsets to ensure a disjoint distribution of speakers and synthesis algorithms in training and test sets and allow open-set analysis.

D. Training setup

We trained and tested the two considered models using the same setup across all the analyzed features, to guarantee the fairest condition possible in their evaluation. We trained the models for 100 epochs using a batch size of 64 samples. We used focal loss with $\gamma = 2$ to help the model focus on learning from difficult samples. The learning rate followed a cosine annealing schedule, starting at 10^{-4} and gradually decaying to 10^{-7} over the course of training. Additionally, a weight decay of 10^{-4} was applied to regularize the process, and the best-performing model was selected based on the lowest validation loss value. Due to the significant class imbalance in the two datasets, where deepfake samples far outnumber bonafide ones, we applied random oversampling during training to ensure that each batch contained an equal number of samples from both classes.

IV. Results

In this section, we evaluate the performance of the considered models, comparing the use of 10 different audio feature sets for detecting both singing voice and speech deepfakes. We aim to determine if some feature sets are more effective than others and explore whether their effectiveness varies between detecting singing voice and speech deepfakes.

As a first experiment, we evaluate the two CNN-based detectors we considered, i.e., ResNet-18 and LCNN, on the SVDD task, which is the primary focus of this paper. Figure 2 shows the results on the CtrSVDD dataset, showing the Area Under the Curve (AUC) values for each feature set considered. The figure shows that, on average, ResNet-18 slightly outperforms the LCNN model. However, in most cases, the performance of both models is quite similar, suggesting that the achieved results are more influenced by the considered audio features rather than the architecture of the models themself. Specifically, log-spectrograms perform the best among handcrafted features, while MFCCs and LFCCs yield poorer detection results. When we consider learned features, such as LEAF and SincConv, ResNet-18 delivers better performance than LCNN. Notably, Wav2Vec 2.0 outperforms all other feature sets, achieving an AUC value close to 95% with both detectors.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on February 28,2026 at 09:53:11 UTC from IEEE Xplore. Restrictions apply.

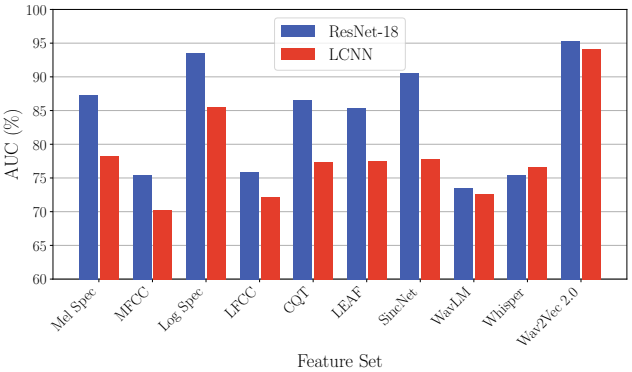

Fig. 2. Performance comparison between ResNet-18 and LCNN models on the CtrSVDD dataset across various feature sets.

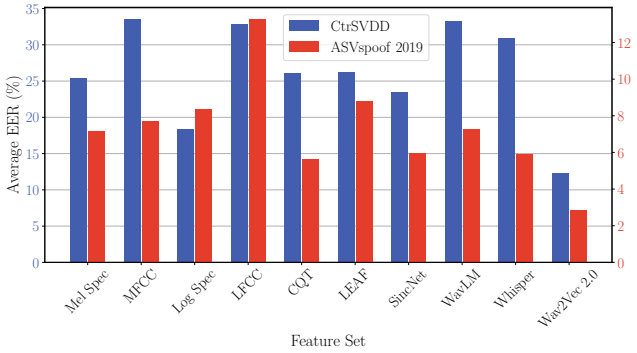

Fig. 3. Average EER using different feature sets on CtrSVDD and ASVspoof2019. Red (right) axis for ASVspoof2019, blue (left) axis for CtrSVDD. Vertical axis direction is reversed for comparison with Figure 2.

TABLE I
EER SCORES OF RESNET-18 AND LCNN MODELS WITH DIVERSE AUDIO FEATURES AS INPUT ON THE CTRSVDD AND ASVSPOOF 2019 DATASETS.

	CtrSVDD		ASVspoof 2019	
	ResNet-18	LCNN	ResNet-18	LCNN
Mel Spec	20.73%	30.00%	7.42%	6.87%
MFCC	31.11%	35.87%	6.52%	8.89%
Log Spec	14.52%	22.46%	5.67%	11.04%
LFCC	31.39%	34.21%	15.35%	11.16%
CQT	21.83%	30.30%	4.91%	6.38%
LEAF	23.08%	29.49%	8.54%	9.12%
SincConv	17.37%	29.66%	6.12%	5.85%
WavLM	33.10%	33.39%	7.24%	7.35%
Whisper	31.29%	30.62%	5.59%	6.19%
Wav2Vec 2.0	11.72%	12.78%	2.07%	3.63%

We now expand our analysis to include results from both singing voice and speech deepfake detection tasks. Table I shows the Equal Error Rate (EER) scores for all considered models on the CtrSVDD and ASVspoof 2019 datasets. Best-performing model scores are highlighted in green, while the worst-performing ones are marked in red.

The results show a significant improvement in classification accuracy on ASVspoof 2019 compared to CtrSVDD, for all the considered features. This discrepancy may be attributed to the fact that the CtrSVDD dataset includes singing voice generation algorithms in the evaluation set (i.e., A12 and A14) that significantly diverge from the training distribution, leading to poorer performance, as noted in the description of the dataset of the SVDD 2024 challenge [34].

Figure 3 presents the EER values for each input feature set across the two datasets, averaging the scores from ResNet-18 and LCNN to minimize model dependency. The results for each dataset are plotted on separate scales to facilitate comparison and reduce the impact of the datasets themself on the observed values. There is a noticeable difference between the scores achieved by the same feature sets when changing the task. Mel-scale-based features, such as mel spectrograms and MFCCs, have good performance in speech deepfake detection but are less effective for singing voice deepfake detection. This might be because mel features emphasize low frequencies, which are relevant for analyzing speech but less applicable for singing voice signals. In contrast, Log Spectrograms yield relatively more modest results on speech but excel in singing voice detection. This is likely because Log Spec features provide a more detailed representation of the entire frequency spectrum, capturing the nuances and harmonic richness characteristic of singing voices. Finally, Wav2Vec 2.0 features deliver the best results for both tasks, reinforcing the growing use of these pre-trained representations in state-of-the-art applications.

V. CONCLUSIONS

In this paper, we investigated the effectiveness of diverse audio feature sets for the SVDD task. By considering 2 CNN-based models and 10 different audio representations, we conducted a comprehensive evaluation of their performance on both singing voice and speech deepfake detection tasks, yielding several insights.

Our findings indicate that feature effectiveness varies significantly with the task. A feature set effective for speech deepfake detection does not necessarily achieve similar results for singing voice detection, and vice versa. For example, Mel scale-based features, which emphasize characteristics well-suited for speech processing, underperform when applied to singing voice detection. In contrast, log spectrogram features, which are more effective at capturing the overall frequency content across the entire spectrum, excel in singing voice analysis but perform less effectively for plain speech. This underscores the need for task-specific feature selection. Finally, Wav2Vec 2.0 representations consistently outperformed other feature sets for both tasks, indicating the value of pre-trained models in capturing complex audio patterns that are valuable across different domains.

Future works will explore further improvements in feature extraction techniques and study the combination of multiple feature sets. While certain features outperform others, each set captures unique aspects that others might overlook, even if those features are less effective overall. Therefore, exploring effective methods for combining features could lead to the development of novel systems tailored to enhance detection capabilities in SVDD.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on February 28,2026 at 09:53:11 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] J. Brodkin. (2024, sep) FBI busts musicians' elaborate AI-powered $10M streaming royalty heist. Accessed: 2024-09-11. [Online]. Available: https://arstechnica.com/information-technology/2024/09/fbi-busts-musicians-elaborate-ai-powered-10m-streaming-royalty-heist/

[2] H. H. S. Josan, "AI and Deepfake Voice Cloning: Innovation, Copyright and Artists' Rights," *Artificial Intelligence*, 2024.

[3] Y. Xie, J. Zhou, X. Lu, Z. Jiang, Y. Yang, H. Cheng, and L. Ye, "FSD: An initial chinese dataset for fake song detection," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[4] E. Conti, D. Salvi, C. Borrelli, B. Hosler, P. Bestagini, F. Antonacci, A. Sarti, M. C. Stamm, and S. Tubaro, "Deepfake speech detection through emotion recognition: a semantic approach," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.

[5] L. Attorresi, D. Salvi, C. Borrelli, P. Bestagini, and S. Tubaro, "Combining Automatic Speaker Verification and Prosody Analysis for Synthetic Speech Detection," in *International Conference on Pattern Recognition (ICPR)*, 2022.

[6] P. Kawa, M. Plata, M. Czuba, P. Szymanski, and P. Syga, "Improved DeepFake Detection Using Whisper Features," in *Interspeech*, 2023.

[7] Y. Guo, H. Huang, X. Chen, H. Zhao, and Y. Wang, "Audio Deepfake Detection With Self-Supervised Wavlm And Multi-Fusion Attentive Classifier," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[8] A. K. S. Yadav, K. Bhagtani, D. Salvi, P. Bestagini, and E. J. Delp, "Fairssd: Understanding bias in synthetic speech detectors," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop (CVPRW)*, 2024.

[9] X. Chen, H. Wu, J.-S. R. Jang, and H.-y. Lee, "Singing voice graph modeling for singfake detection," in *Interspeech*, 2024.

[10] Y. Zang, Y. Zhang, M. Heydari, and Z. Duan, "SingFake: Singing voice deepfake detection," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[11] J.-w. Jung, H.-S. Heo, H. Tak, H.-j. Shim, J. S. Chung, B.-J. Lee, H.-J. Yu, and N. Evans, "AASIST: Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention Networks," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.

[12] Y. Zhang, F. Jiang, and Z. Duan, "One-class learning towards synthetic voice spoofing detection," *IEEE Signal Processing Letters*, vol. 28, pp. 937–941, 2021.

[13] H. Tak, M. Todisco, X. Wang, J.-W. Jung, J. Yamagishi, and N. Evans, "Automatic speaker verification spoofing and deepfake detection using wav2vec 2.0 and data augmentation," in *Interspeech*, 2022.

[14] Y. Yang, H. Qin, H. Zhou, C. Wang, T. Guo, K. Han, and Y. Wang, "A robust audio deepfake detection system via multi-view feature," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[15] N. M. Müller, P. Czempin, F. Dieckmann, A. Froghyar, and K. Böttinger, "Does Audio Deepfake Detection Generalize?" in *Interspeech*, 2022.

[16] D. Salvi, P. Bestagini, and S. Tubaro, "Towards frequency band explainability in synthetic speech detection," in *European Signal Processing Conference (EUSIPCO)*, 2023.

[17] J. Liu, C. Li, Y. Ren, F. Chen, and Z. Zhao, "Diffsinger: Singing voice synthesis via shallow diffusion mechanism," in *AAAI Conference on Artificial Intelligence*, 2022.

[18] Y. Zhang, R. Huang, R. Li, J. He, Y. Xia, F. Chen, X. Duan, B. Huai, and Z. Zhao, "StyleSinger: Style Transfer for Out-of-Domain Singing Voice Synthesis," in *AAAI Conference on Artificial Intelligence*, 2024.

[19] H. Tak, J. Patino, M. Todisco, A. Nautsch, N. Evans, and A. Larcher, "End-to-end anti-spoofing with RawNet2," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021.

[20] N. Zeghidour, O. Teboul, F. D. C. Quitry, and M. Tagliasacchi, "Leaf: A learnable frontend for audio classification," *arXiv preprint arXiv:2101.08596*, 2021.

[21] M. Ravanelli and Y. Bengio, "Speaker recognition from raw waveform with sincnet," in *IEEE Spoken Language Technology Workshop (SLT)*, 2018.

[22] E. Goron, L. Asai, E. Rut, and M. Dinov, "Improving Domain Generalization in Speech Emotion Recognition with Whisper," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[23] R. Gupta, C. Madill, D. R. Gunjawate, D. D. Nguyen, and C. T. Jin, "Addressing Data Scarcity in Voice Disorder Detection with Self-Supervised Models," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[24] D. Salvi, A. K. S. Yadav, K. Bhagtani, V. Negroni, P. Bestagini, and E. J. Delp, "Comparative Analysis of ASR Methods for Speech Deepfake Detection," in *Asilomar Conference on Signals, Systems, and Computers*, 2024.

[25] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *International Conference on Machine Learning (ICML)*. PMLR, 2023.

[26] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao *et al.*, "Wavlm: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.

[27] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," *Advances in neural information processing systems*, vol. 33, pp. 12 449–12 460, 2020.

[28] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[29] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009.

[30] C. Borrelli, P. Bestagini, F. Antonacci, A. Sarti, and S. Tubaro, "Synthetic speech detection through short-term and long-term prediction traces," *EURASIP Journal on Information Security*, vol. 2021, pp. 1–14, 2021.

[31] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: a simple way to prevent neural networks from overfitting," *The journal of machine learning research*, vol. 15, no. 1, pp. 1929–1958, 2014.

[32] X. Wu, R. He, Z. Sun, and T. Tan, "A light CNN for deep face representation with noisy labels," *IEEE Transactions on Information Forensics and Security*, vol. 13, no. 11, pp. 2884–2896, 2018.

[33] X. Liu, X. Wang, M. Sahidullah, J. Patino, H. Delgado, T. Kinnunen, M. Todisco, J. Yamagishi, N. Evans, A. Nautsch *et al.*, "ASVspoof 2021: Towards spoofed and deepfake speech detection in the wild," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 2507–2522, 2023.

[34] Y. Zang, J. Shi, Y. Zhang, R. Yamamoto, J. Han, Y. Tang, S. Xu, W. Zhao, J. Guo, T. Toda, and Z. Duan, "Ctrsvdd: A benchmark dataset and baseline analysis for controlled singing voice deepfake detection," in *Interspeech*, 2024.

[35] Y. Wang, X. Wang, P. Zhu, J. Wu, H. Li, H. Xue, Y. Zhang, L. Xie, and M. Bi, "Opencpop: A high-quality open source chinese popular song corpus for singing voice synthesis," *arXiv preprint arXiv:2201.07429*, 2022.

[36] L. Zhang, R. Li, S. Wang, L. Deng, J. Liu, Y. Ren, J. He, R. Huang, J. Zhu, X. Chen *et al.*, "M4singer: A multi-style, multi-singer and musical score provided mandarin singing corpus," *Advances in Neural Information Processing Systems*, vol. 35, pp. 6914–6926, 2022.

[37] J. Shi, Y. Lin, X. Bai, K. Zhang, Y. Wu, Y. Tang, Y. Yu, Q. Jin, and S. Watanabe, "Singing voice data scaling-up: An introduction to ace-opencpop and kising-v2," *arXiv preprint arXiv:2401.17619*, 2024.

[38] I. Ogawa and M. Morise, "Tohoku kiritan singing database: A singing database for statistical parametric singing synthesis using japanese pop songs," *Acoustical Science and Technology*, 2021.

[39] H. Tamaru, S. Takamichi, N. Tanji, and H. Saruwatari, "Jvs-music: Japanese multispeaker singing-voice corpus," *arXiv preprint arXiv:2001.07044*, 2020.

[40] M. Todisco, X. Wang, V. Vestman, M. Sahidullah, H. Delgado, A. Nautsch, J. Yamagishi, N. Evans, T. Kinnunen, and K. A. Lee, "ASVspoof 2019: Future horizons in spoofed and fake audio detection," in *Interspeech*, 2019.

[41] C. Veaux, J. Yamagishi, K. MacDonald *et al.*, "Superseded-CSTR VCTK Corpus: English Multi-Speaker Corpus for CSTR Voice Cloning Toolkit," *University of Edinburgh. The Centre for Speech Technology Research (CSTR)*, 2016.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on February 28,2026 at 09:53:11 UTC from IEEE Xplore. Restrictions apply.